Paper Title

Author Name

***Abstract*—**

***Index Terms*—**

I. INTRODUCTION

Deep learning has achieved strong performance in many radar applications, including target detection, classification, and recognition. Its performance, however, depends strongly on the quantity, diversity, and annotation quality of the available data. This dependence is particularly restrictive for maritime radar because existing measured datasets cover only a limited set of sea states, radar parameters, viewing geometries, and target conditions. A model trained on such data may not generalize to an unobserved wind speed, wave direction, grazing angle, or clutter regime. Expanding a measured dataset is also difficult: sea experiments require suitable platforms, radar equipment, environmental sensors, synchronized observations, and favorable weather, while a desired sea condition cannot be reproduced on demand. The shortage and limited diversity of measured radar data have therefore become a major constraint on the development of data-driven maritime sensing methods.

Numerical simulation provides a practical way to supplement measured data because environmental and radar parameters can be controlled independently and a large number of labeled samples can be generated repeatedly. For radar applications, however, computational efficiency alone is insufficient. A simulated sea surface must preserve the statistical roughness of the background sea, respond reasonably to different wind and wave conditions, and represent the local structures that produce strong or non-Gaussian radar returns. High-fidelity hydrodynamic methods can resolve nonlinear wave evolution and breaking interfaces in detail, while full-wave electromagnetic methods can describe the associated diffraction and multipath interactions. Their volume-mesh evolution, fine spatial discretization, or global electromagnetic interactions nevertheless impose substantial computational and storage costs, making repeated simulation over many environmental and radar conditions difficult. Spectral sea-surface and facet-based scattering models are considerably more efficient, but a conventional linear realization does not explicitly represent strongly nonlinear or overturning crests. A useful data-generation method must therefore balance physical credibility, environmental adaptability, computational cost, and compatibility with radar scattering calculations.

Sea-surface roughness is inherently multiscale. Long gravity waves determine the large-scale tilt and motion of the surface, whereas shorter wind waves modify local slopes and radar-resonant components. Consequently, the observed return depends jointly on sea state, incidence angle, radar wavelength, and surface geometry. High-resolution measurements show that sea clutter may exhibit pronounced non-Gaussian tails and intermittent sea spikes, which can be confused with target responses [1]–[5]. Directional wave spectra, such as the unified Elfouhaily spectrum, provide an efficient description of the gravity and gravity–capillary energy distribution [6]. Nevertheless, a linear spectral realization is approximately Gaussian and mainly preserves second-order statistics. Nonlinear hydrodynamic interactions sharpen crests, smooth troughs, and change the distributions of slopes and curvatures, thereby influencing the scattering coefficient and Doppler spectrum [7]. The sea-surface skewness function and its small-scale-wave parameter also affect the upwind–downwind asymmetry of radar backscattering [8], while convolutions of short-wave spectral components can contribute appreciably to the normalized backscattering cross section under different radar frequencies, wind speeds, and incidence angles [9]. Thus, a radar-oriented sea model should represent not only the total spectral energy but also nonlinear crest geometry and the directional organization of shorter wind waves.

The return from an ordinary rough sea surface is commonly described by quasi-specular, Bragg, or two-scale scattering formulations. Their relative contributions vary with incidence angle, and both quasi-specular and Bragg mechanisms may be required in the transition range [10]. Radar observations of marine film slicks further suggest that the source of short wind waves can depend on propagation direction and cannot always be represented by conventional wind-input terms alone [11]. For steep and breaking waves, additional scattering mechanisms become important. Numerical and experimental studies have related low-grazing-angle sea spikes to steepening or breaking crests [12]–[14]. Full-wave and hybrid models have investigated crest diffraction, rough-face scattering, and multipath interactions [15]–[19]. Facet-based methods have introduced breaking-wave contributions into larger sea surfaces and analyzed their effects on normalized radar cross section (NRCS), sea spikes, and synthetic aperture radar images [20]. Three-dimensional plunging-breaker models combined with ray tracing have also shown that a breaker crest can generate target-like peaks in radar cross section and high-resolution range profiles [21].

Existing approaches present two complementary limitations. On the electromagnetic side, rigorous or high-order numerical methods can resolve diffraction, rough-face scattering, and multipath interactions from breaking waves, but they require fine discretization and repeated solution of large scattering systems [15], [19]. At X-band, conventional numerical discretization may require surface samples much smaller than the radar wavelength, making time-evolving and large-area calculations particularly expensive [7]; even accelerated surface-scattering methods report substantial costs for multiview or full-polarimetric calculations [22]. Lower-cost electromagnetic approximations also have restricted validity. The traditional two-scale model depends on a prescribed separation between quasi-specular and Bragg regimes and loses accuracy where the two mechanisms coexist [10], whereas steep or multi-valued breaker faces introduce crest diffraction, shadowing, and multipath contributions that are not represented by an ordinary rough-surface term alone [14], [17], [18]. Thus, electromagnetic fidelity and computational efficiency remain difficult to obtain simultaneously for large-scale radar-data generation.

On the sea-surface side, computationally efficient spectral and replacement-based methods have different limitations. Linear spectral synthesis preserves the prescribed second-order spectrum but cannot reproduce crest–trough asymmetry or overturning geometry, while Creamer- or Lie-type nonlinear surfaces improve slopes and higher-order statistics but generally remain single-valued height fields [7], [8]. Existing breaking-wave studies often use a selected LONGTANK stage, a measured breaker profile, or a prescribed analytical three-dimensional breaker and then insert or superimpose that geometry into the scattering scene [12], [16], [20], [21]. These constructions are effective for investigating the radar signature of a particular breaker, but their emphasis is not the automatic generation of diverse surfaces conditioned jointly on wind speed, wind direction, short-wave organization, and local crest state [20], [21]. This limitation is important because radar observations indicate that short-wave generation and damping can depend strongly on propagation direction and surface condition [11]. Directly superimposing an independently generated short-wave field without band separation or renormalization can duplicate spectral contributions already contained in the background surface [7], [9], while replacing a region with a prescribed breaker profile may limit geometric continuity and adaptability across sea states [12], [20], [21]. A rapid dataset-oriented model should therefore connect a statistically constrained background, directionally organized short waves, and locally triggered curling geometry within the same environmental parameterization. Its advantage should be evaluated jointly in terms of physical representativeness, environmental flexibility, radar applicability, and measured computational scalability.

This letter proposes a hierarchical sea-surface simulation method intended for rapid radar-data generation under variable environmental conditions. First, an Elfouhaily directional spectrum and a wind-dependent nonlinear Lie transformation are used to construct a statistically constrained nonlinear background surface. Second, a selected short-wave band is replaced by an equal-energy directional component, forming shorter wind-wave ridges with a prescribed dominant propagation direction without repeatedly injecting spectral energy. Third, candidate crests are identified from elevation, propagation-direction slope, and curvature, after which a compactly supported coordinate transformation generates a localized forward-extending and downward-curling breaker. Wind speed, wind direction, directional spreading, short-wave scale, and local curling parameters can be varied to produce different sea-surface realizations, while the facet representation allows the generated surfaces to be used directly in radar scattering and echo simulations. Because the nonlinear and curling operations are applied through spectral and sparse local surface updates, the method avoids repeated three-dimensional fluid evolution and dense full-wave interaction for every generated realization. It is therefore designed to retain radar-relevant breaker geometry while remaining suitable for large, controllable synthetic radar datasets.

The method is evaluated from surface geometry to radar observables and computational scalability. The background and short-wave-enriched surfaces are first examined using spectral consistency and along-wind and crosswind mean-square slopes under different wind conditions. The localized breaker is then assessed using crest geometry, facet-orientation statistics, radar-facing projected area, and paired unit-footprint scattering enhancement. Radar echoes generated from the complete surface are evaluated through range–time intensity, temporal correlation, and amplitude-distribution statistics. Finally, the measured runtime is decomposed into surface generation, facet-level scattering, echo formation, and input/output costs; its scaling with surface resolution and dataset size is compared with reported X-band numerical methods, with differences in output scope stated explicitly [22]–[24]. Section II presents the hierarchical surface model. Section III describes the validation metrics and radar echo generation. Section IV reports the surface and radar comparisons. Section V analyzes computational cost and dataset scalability, and Section VI concludes this letter.

II. Hierarchical Sea-Surface Modeling

The proposed framework maps environmental conditions and random realizations to radar-ready dynamic sea surfaces. The environmental input is $\boldsymbol{p}_e = \{U_{10}, \phi_w, \Omega_c, H_s, L_x, L_y, \Delta x, \Delta y\}$, where U_{10}, ϕ_w, Ω_c, and H_s specify the sea state, while the remaining quantities define the computational domain. A random variable set $\boldsymbol{\xi}$ controls the Fourier phases, allowing independent samples to be generated under identical environmental conditions. Radar parameters are introduced only after the surface has been triangulated, so the sea state and observation configuration can be varied independently.

The complete generation chain is summarized as

$$\begin{aligned}(\boldsymbol{p}_e, \boldsymbol{\xi}) &\xrightarrow{\mathcal{G}_{NL}} \eta_N \xrightarrow{\mathcal{D}_s} \eta_D \\ &\xrightarrow{\mathcal{C}} \mathcal{M}_C \xrightarrow{\mathcal{R}(\boldsymbol{p}_r)} \boldsymbol{y},\end{aligned} \tag{1}$$

where $\mathcal{G}_{NL}$ denotes nonlinear spectral background generation, $\mathcal{D}_s$ denotes equal-energy directional short-wave replacement, $\mathcal{C}$ denotes localized curling and triangulation, and $\mathcal{R}$ denotes facet-based radar simulation. The first two operators act over the full domain using Fourier transforms, whereas the curling operator is restricted to selected steep crests. This separation retains large-area statistical control while limiting the additional geometric computation to radar-relevant local structures.

A. Nonlinear Background and Directional Wind Waves

The first modeling stage combines a nonlinear broadband background with an equal-energy directional reorganization of

Placeholder for the overall framework figure

Environmental parameters and random phases $\rightarrow$ Elfouhaily spectral sampling and nonlinear Lie transformation $\rightarrow$ equal-energy directional short-wave replacement $\rightarrow$ steep-crest detection and localized curling $\rightarrow$ triangular facets and radar echoes.

Fig. 1. Overall framework of the proposed sea-surface and radar-echo simulation method.

its resolved short-wave band. The background supplies sea-state-dependent spectral statistics and non-Gaussian crest geometry, while the reorganized band produces short wind-wave ridges with a controllable dominant propagation direction without duplicating energy already present in the Elfouhaily spectrum.

Let $k = (k_x^2 + k_y^2)^{1/2}$ and $\varphi = \operatorname{atan2}(k_y, k_x)$. The directional elevation spectrum is written as [6]

$$\Psi_\eta(k,\varphi) = \frac{B_l(k) + B_h(k)}{k^4} D(k, \varphi - \varphi_w), \tag{2}$$

where B_l and B_h are the gravity- and capillary-wave curvature spectra and D is the Elfouhaily spreading function. On the discrete wavenumber grid, the linear coefficients are sampled as

$$\widehat{\eta}_L(\boldsymbol{k}) = \left[\Psi_\eta(\boldsymbol{k}) \Delta k_x \Delta k_y\right]^{1/2} \xi(\boldsymbol{k}), \tag{3}$$

where ξ is a unit-variance complex Gaussian field. Hermitian symmetry is imposed explicitly, and the same random phases are retained during the nonlinear transformation.

To express the Lie correction relative to the wind, the along- and cross-wind wavenumbers are defined as $k_a = k_x \cos\varphi_w + k_y \sin\varphi_w$ and $k_c = -k_x \sin\varphi_w + k_y \cos\varphi_w$. The corresponding Riesz fields are

$$\widehat{h}_a = -\mathrm{i}\frac{k_a}{k}\widehat{\eta}_L \mathcal{B}_i, \qquad \widehat{h}_c = -\mathrm{i}\frac{k_c}{k}\widehat{\eta}_L \mathcal{B}_i, \tag{4}$$

where $\mathcal{B}_i$ is the input-band mask. Following the second-order Lie/Creamer construction [7], the nonlinear correction is

$$\begin{aligned} \widehat{L} = & -\frac{k_a^2}{2k}\mathcal{F}\{h_a^2\} - \frac{k_a k_c}{k}\mathcal{F}\{h_a h_c\} \\ & -\frac{k_c^2}{2k}\mathcal{F}\{h_c^2\}, \end{aligned} \tag{5}$$

and the preliminary nonlinear spectrum becomes

$$\widehat{\eta}_{N,0} = \widehat{\eta}_L + \beta \mathcal{P}_H \left\{ \mathcal{B}_o \widehat{L} \right\}, \tag{6}$$

where β is dimensionless, $\mathcal{B}_o$ is the output-band mask, and $\mathcal{P}_H$ denotes Hermitian projection. The Riesz multipliers and β are dimensionless; hence, no dimensional wind-speed factor is directly multiplied by the elevation correction. The input and output limits are set to $4k_p$ and $8k_p$, respectively, to retain the resolved quadratic interactions.

Because the nonlinear operation redistributes slope energy, its along- and cross-wind mean-square slopes (MSSs) are closed against the same Elfouhaily spectrum used for synthesis. For $q \in \{a, c\}$,

$$\begin{aligned} m_q &= \sum_{\boldsymbol{k}} k_q^2 \left|\widehat{\eta}(\boldsymbol{k})\right|^2, \\ G(\varphi) &= \exp\left[\frac{\lambda_0 + \lambda_2 \cos 2(\varphi - \varphi_w)}{2}\right]. \end{aligned} \tag{7}$$

Independent calibration realizations determine smooth wind-dependent along- and cross-wind scale factors by matching the median raw nonlinear MSSs to the integrated Elfouhaily MSSs. For each subsequent realization, λ_0 and λ_2 are solved to realize these fixed scale factors without altering Fourier phase. Cox–Munk and the other empirical slope models are therefore reserved for external validation rather than used as fitting constraints. The closed nonlinear background spectrum is $\widehat{\eta}_N = G(\varphi)\widehat{\eta}_{N,0}$.

The broadband background already contains short-wave energy, but it does not ensure an organized local propagation direction. A target wavelength interval $[\lambda_{s,\min}, \lambda_{s,\max}]$ is therefore selected, with $k_l = 2\pi/\lambda_{s,\max}$ and $k_h = 2\pi/\lambda_{s,\min}$. Let $W_s(k)$ be a raised-cosine radial window over this interval. The retained background is

$$\widehat{\eta}_r(\boldsymbol{k}) = \left[1 - W_s(k)\right] \widehat{\eta}_N(\boldsymbol{k}). \tag{8}$$

For a prescribed propagation unit vector $\boldsymbol{e}_s = (\cos\psi_s, \sin\psi_s)$, the replacement-band shape is

$$\Phi_s(\boldsymbol{k}) = W_s(k) \exp\left[-\frac{\log^2(k/k_0)}{2\sigma_k^2} - \frac{\theta_a^2}{2\sigma_\theta^2}\right], \tag{9}$$

where $k_0 = \sqrt{k_l k_h}$, $\sigma_k = \log(k_h/k_l)/4$, and $\theta_a = \cos^{-1}(|\boldsymbol{k} \cdot \boldsymbol{e}_s|/k)$. The angular term concentrates the band around the prescribed axis and represents the directional organization observed for short wind waves [11].

A preliminary replacement spectrum is sampled as $\widehat{\eta}_{s,0} = \mathcal{P}_H\{\sqrt{\Phi_s}\, \xi_s\}$. Its scale is determined from the removed-band energy:

$$\alpha_s = \left[\rho_s \frac{\sum_{\boldsymbol{k}} |W_s \widehat{\eta}_N|^2}{\sum_{\boldsymbol{k}} |\widehat{\eta}_{s,0}|^2}\right]^{1/2}. \tag{10}$$

With $\rho_s = 1$, the replacement and removed bands have equal variance by Parseval's theorem. This is a spectral reorganization rather than an additive energy injection.

Placeholder for the linear Elfouhaily surface, nonlinear Lie background, removed short-wave band, equal-energy directional replacement, and combined surface.

Fig. 2. Construction of the nonlinear background with directionally organized short wind waves.

Unidirectional evolution is introduced by assigning opposite temporal phases to the Hermitian wavenumber pair. The combined surface is

$$\eta_D(\boldsymbol{x},t) = \operatorname{Re}\mathcal{F}^{-1}\Big\{\widehat{\eta}_r(\boldsymbol{k})\mathrm{e}^{-\mathrm{i}\omega(k)t} + \alpha_s\widehat{\eta}_{s,0}(\boldsymbol{k})\mathrm{e}^{-\mathrm{i}\omega(k)\,\mathrm{sgn}(\boldsymbol{k}\cdot\boldsymbol{e}_s)t}\Big\}, \quad (11)$$

where $\omega(k) = \sqrt{gk}$ for the resolved short gravity waves. Unless otherwise stated, $\lambda_{s,\min} = 1.5$ m, $\lambda_{s,\max} = 5.0$ m, $\sigma_\theta = 8^\circ$, and $\rho_s = 1$. The minimum wavelength is represented by at least six grid samples. The resulting η_D retains the nonlinear broadband background while providing directionally organized short crests for the local curling stage.

B. Localized Crest Detection and Curling Geometry

The nonlinear directional surface η_D remains single valued and therefore cannot represent a forward-overturning crest. A localized parametric transformation is consequently applied only to selected steep crests. The construction is intended to retain radar-relevant upper-surface geometry at substantially lower cost than resolving the complete air–water flow; it does not model entrained air, spray, or the full hydrodynamic evolution of breaking waves. Similar finite breaker geometries have been used to study localized electromagnetic signatures [12], [20], [21].

Let $\boldsymbol{e}_s = (\cos\psi_s, \sin\psi_s)$ denote the dominant propagation direction and $\partial_u = \boldsymbol{e}_s\cdot\nabla$. Candidate crests are restricted to the set

$$\mathcal{D}_c = \left\{\boldsymbol{x} : \eta_D \ge h_q,\ \partial_u^2\eta_D < 0,\ |\partial_u\eta_D| \le s_c\right\}, \quad (12)$$

where h_q is a prescribed elevation quantile and s_c is a near-crest slope tolerance. Edge points that cannot contain the complete curling support are excluded. Within $\mathcal{D}_c$, the event center is selected using

$$\boldsymbol{x}_c = \underset{\boldsymbol{x}\in\mathcal{D}_c}{\arg\max}\, Q(\boldsymbol{x}), \qquad Q = w_h\widetilde{\eta} + w_\kappa\widetilde{(-\partial_u^2\eta_D)} + w_s\widetilde{\eta}_s, \quad (13)$$

where tildes denote normalized positive quantities and η_s is the directional short-wave component. This rule favors high crests with negative directional curvature and a locally strong organized short-wave contribution. It is used to place a parameterized breaking event and is not presented as a universal hydrodynamic breaking-onset criterion.

At $\boldsymbol{x}_c = (x_c, y_c)$, propagation and crestwise coordinates are defined as

$$\begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} \cos\psi_s & \sin\psi_s \\ -\sin\psi_s & \cos\psi_s \end{bmatrix}\begin{bmatrix} x - x_c \\ y - y_c \end{bmatrix}. \quad (14)$$

Let $a = L_c/2$ and $b = W_c/2$ be the crestwise half-length and propagation-direction half-width. A compact cosine window is defined by

$$C(q) = \begin{cases} \frac{1}{2}[1 + \cos(\pi q)], & |q| < 1, \\ 0, & |q| \ge 1. \end{cases} \quad (15)$$

To avoid a perfectly regular footprint, a small centerline displacement $u_c(v)$ and width factor $r(v)$ are introduced and $u_r = u - u_c(v)$ is used below. The core and transition supports are

$$W_c = C\left(\frac{u_r}{br(v)}\right)C\left(\frac{v}{a}\right),$$
$$W_t = C\left(\frac{u_r}{(b+\ell_t)r(v)}\right)C\left(\frac{v}{a}\right), \quad (16)$$

where $\ell_t > 0$ extends the transition region beyond the rotation core. Both windows vanish continuously at their boundaries, while W_t permits the neighboring non-overturned surface to evolve smoothly before returning to the unchanged background.

The approaching crest is first modified without rotation. An asymmetric width

$$\sigma(u_r) = \begin{cases} \sigma_r, & u_r < 0, \\ \sigma_f, & u_r \ge 0, \end{cases} \qquad \sigma_f < \sigma_r, \quad (17)$$

produces a broader rear face and a steeper forward face. With σ_e denoting the transition scale, the pre-rotation coordinates are

$$z_p = \eta_D + A_c\exp\left[-\frac{u_r^2}{2\sigma^2(u_r)}\right]W_t + A_e\exp\left[-\frac{u_r^2}{2\sigma_e^2}\right]W_t,$$
$$u_p = u + A_l\exp\left[-\frac{u_r^2}{2\sigma_e^2}\right]W_t. \quad (18)$$

Here, A_c controls the local crest correction, whereas A_e and A_l produce the low-amplitude elevation and forward-lean changes across the wider evolution band. This separation prevents the curl from appearing as an isolated bump placed directly at the highest point.

The rotation center is shifted by $d_c < 0$ toward the approaching shoulder, and the local angle field is

$$\theta(u,v) = \theta_{\max}\exp\left[-\frac{(u_r - d_c)^2}{2(0.62b)^2}\right]C\left(\frac{v}{a}\right). \quad (19)$$

The rotation axis is parallel to the crest direction, with pivot coordinates $u_0 = u_c(v) + d_c + d_f$ and $z_0 = \eta_D - d_p$, where d_f is the forward offset and d_p is the pivot depth. Defining $\Delta u = u_p - u_0$ and $\Delta z = z_p - z_0$, the rotated coordinates are

$$\begin{bmatrix} u_R \\ z_R \end{bmatrix} = \begin{bmatrix} u_0 \\ z_0 \end{bmatrix} + \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}\begin{bmatrix} \Delta u \\ \Delta z \end{bmatrix}. \quad (20)$$

Placeholder for crest candidates and selected center, core and transition supports, propagation-direction profile before and after curling, and the final triangulated overturning patch.

Fig. 3. Localized curling-wave construction on a selected directional crest. The rotation begins on the approaching shoulder and is blended into the background through a wider transition region.

The final local coordinates are blended with the wider transition support:

$$\begin{aligned} &u' = u_p + W_t(u_R - u_p), \qquad z' = z_p + W_t(z_R - z_p), \\ &v' = v. \end{aligned} \tag{21}$$

They are mapped back to the global frame by $x' = x_c + u' \cos\psi_s - v' \sin\psi_s$ and $y' = y_c + u' \sin\psi_s + v' \cos\psi_s$. Because the output is the parametric surface $\boldsymbol{r}_C(u, v) = [x', y', z']^{\mathrm{T}}$ rather than a height function $z = \eta(x, y)$, the geometry can represent locally overlapping horizontal projections.

The core event mask and the actual overturning mask are distinguished as

$$\begin{aligned} &M_c = \mathbf{1}(W_c \geq \tau_c), \qquad M_o = M_c\,\mathbf{1}(J_u < 0), \\ &J_u = \boldsymbol{e}_s \cdot \nabla u'. \end{aligned} \tag{22}$$

M_c identifies facets materially affected by the curling core, whereas $J_u < 0$ identifies a reversal of the propagation-coordinate mapping and therefore provides a direct geometric indication of overturning. The transition support W_t is not counted as the breaking core.

Finally, the original structured-grid connectivity is retained to triangulate both the uncurled and curled surfaces. For each face $f = (i, j, k)$, its area and unit normal are

$$\begin{aligned} &\boldsymbol{e}_{ij} = \boldsymbol{r}_j - \boldsymbol{r}_i, \qquad \boldsymbol{e}_{ik} = \boldsymbol{r}_k - \boldsymbol{r}_i, \\ &A_f = \frac{1}{2}\left\|\boldsymbol{e}_{ij} \times \boldsymbol{e}_{ik}\right\|, \qquad \boldsymbol{n}_f = \frac{\boldsymbol{e}_{ij} \times \boldsymbol{e}_{ik}}{2A_f}. \end{aligned} \tag{23}$$

Preserving the connectivity is essential because retriangulation in the (x', y') projection can incorrectly connect the upper and lower portions of an overturned crest. The paired baseline and curled meshes therefore have identical face indices, allowing changes in local area, orientation, visibility, and radar scattering to be attributed to the curling transformation.

C. Facet-Based Scattering and Radar Echo Generation

The curled surface is converted into radar data through a facet-based mean-power model followed by resolution-cell statistical echo synthesis. The electromagnetic component does not constitute a new full-wave or two-scale scattering theory. Instead, the resolved triangular geometry determines local orientation, projected area, range, antenna weighting, and visibility, while unresolved X-band roughness is represented by the Technology Service Corporation (TSC) empirical sea-reflectivity kernel. This distinction is important because the TSC acronym refers to the empirical model rather than to a "two-scale composite" abbreviation [25].

For facet m, let $\boldsymbol{r}_m$, $\boldsymbol{n}_m$, and A_m denote its center, unit normal, and surface area, and let $\boldsymbol{r}_R$ be the radar position. The slant range and look direction from the facet to the radar are

$$R_m = \|\boldsymbol{r}_R - \boldsymbol{r}_m\|, \qquad \boldsymbol{s}_m = \frac{\boldsymbol{r}_R - \boldsymbol{r}_m}{R_m}. \tag{24}$$

The local grazing angle and relative wind azimuth are

$$\phi_m = \sin^{-1}(\boldsymbol{n}_m \cdot \boldsymbol{s}_m), \qquad \vartheta_m = \mathrm{wrap}_{[-\pi,\pi]}(\varphi_w - \varphi_m), \tag{25}$$

where φ_m is the radar azimuth viewed from the facet. Back-facing facets, facets below the minimum grazing angle, and facets hidden by a larger preceding horizon angle are removed through a binary visibility factor V_m. This update makes the curling transformation affect the return through both facet orientation and self-shadowing.

The conditional mean NRCS is evaluated as the sum of an empirical diffuse term and a quasi-specular term:

$$\sigma_m^0 = \sigma_d^0(\lambda, S, \phi_m, \vartheta_m, U_{10}) + \sigma_q^0(\lambda, S, \phi_m), \tag{26}$$

where S is the Douglas sea-state index and λ is the radar wavelength. In the implemented HH-polarized kernel,

$$\begin{aligned} \sigma_d^0 &= \frac{1.7 \times 10^{-5}\sqrt{\phi_m}}{(3.2808\lambda + 0.05)^{1.8}} F_a F_U F_\vartheta F_p, \\ \sigma_q^0 &= \mu \cot^2\beta \exp\left[-\frac{\cot^2\phi_m}{\tan^2\beta}\right], \end{aligned} \tag{27}$$

with

$$\begin{aligned} \chi &= \frac{4.5416\phi_m(3.2808\sigma_H + 0.25)}{\lambda}, \\ F_a &= \frac{\chi^{1.5}}{1 + \chi^{1.5}}, \qquad F_U = \left(\frac{1.9438U_{10} + 4}{15}\right)^p, \\ F_p &= 1 - 0.6\sin^2\vartheta_m, \qquad F_\vartheta = \exp(\Gamma_\vartheta), \\ \Gamma_\vartheta &= 0.3\cos\vartheta_m \exp(-\phi_m/0.17) \\ &\quad \times (10.7636\lambda^2 + 0.005)^{-0.2}, \end{aligned} \tag{28}$$

where $\sigma_H = 0.03505S^{1.95}$ and p, μ, and β follow the standard wavelength-, grazing-angle-, and sea-state-dependent TSC parameterization. The dB output is converted to linear NRCS before any area weighting. The empirical kernel is used because it provides a low-cost conditional mean for large numbers of facets; physics-based two-scale models remain relevant independent references, particularly in the transition region where Bragg and quasi-specular mechanisms coexist [10].

The horizontal projected area of facet m is $A_{h,m} = A_m \max(\boldsymbol{n}_m \cdot \boldsymbol{e}_z, 0)$, and its monostatic RCS contribution is

$$\sigma_m = V_m \sigma_m^0 A_{h,m}. \tag{29}$$

Let G_m denote the one-way antenna power gain. The mean received power accumulated in range cell b is

$$P_b = \sum_{m \in \mathcal{I}_b} \frac{P_t \lambda^2 G_m^2}{(4\pi)^3 R_m^4} \sigma_m, \tag{30}$$

Placeholder for facet viewing geometry, local grazing angle and visibility, range-bin power accumulation, correlated speckle/texture modulation, pulse compression, and the resulting RTI image.

Fig. 4. Facet-based sea-scattering and radar-echo synthesis chain.

where $\mathcal{I}_b$ contains all visible facets assigned to the cell and P_t is the transmitted peak power. The summation is performed at the conditional-power level rather than by solving mutual electromagnetic interactions among all facets.

Slow-time variability is introduced at the resolution-cell level. A unit-power complex speckle process is formed from fast and slow components,

$$\begin{aligned} s_{n,b} &= \sqrt{1-f_s}\, s_{n,b}^{(f)} + \sqrt{f_s}\, s_{n,b}^{(s)}, \\ s_{n,b}^{(j)} &= \rho_j s_{n-1,b}^{(j)} + \sqrt{1-\rho_j^2}\, w_{n,b}^{(j)}. \end{aligned} \tag{31}$$

where $w_{n,b}^{(j)}$ is a circular complex Gaussian innovation and $\rho_j = \exp[-1/(f_{\mathrm{PRF}}\tau_j)]$. A positive correlated texture $T_{n,b}$ is generated independently and normalized so that $\mathbb{E}[T_{n,b}] = 1$. Thus, texture and speckle alter intermittency and temporal persistence without changing the conditional mean power supplied by the facet model.

Before pulse compression, the complex baseband return is

$$y_{n,b}^{(0)} = \sqrt{P_b T_{n,b}}\, s_{n,b} \exp\left(-\mathrm{i}\frac{4\pi R_b}{\lambda}\right) + n_{n,b}, \tag{32}$$

where $n_{n,b}$ is receiver noise. With the normalized range impulse response h_b, the simulated echo is

$$y_{n,b} = \sum_q h_q y_{n,b-q}^{(0)}. \tag{33}$$

The range–time intensity image is formed from $I_{n,b} = |y_{n,b}|^2$ and displayed using a common reference level across the compared groups. Echo amplitudes or intensities extracted from a predeclared homogeneous range interval are then used for the distribution, tail-probability, temporal-correlation, and sea-spike analyses described in the validation section.

This implementation is a scalable conditional-mean scattering and statistical echo generator, not a coherent per-facet, per-pulse full-wave solver. The TSC kernel does not explicitly model crest diffraction, foam volume scattering, spray, or multipath between overturning faces. Consequently, changes between paired noncurling and curling cases quantify the response produced within the adopted geometry, projected-area, local-angle, visibility, and empirical-reflectivity model; they are not interpreted as the complete electromagnetic response of a real breaking wave.

III. Validation and Results

Validation proceeds from global surface statistics to localized curling-wave geometry and scattering, and finally to full-scene radar signatures and computational efficiency. Accordingly, the evaluation is organized into five levels: large-area slope statistics, local curling-wave geometry, local electromagnetic scattering, full-scene radar echoes, and computational scalability.

A. Global Surface Statistical Validation

This subsection examines whether the generated large-area surfaces retain physically plausible wind-dependent roughness before the localized curling operation is introduced. Three related questions are separated: whether the multiscale linear generator recovers the mean-square slope (MSS) implied by its input Elfouhaily spectrum, whether the MSS-constrained nonlinear surface remains consistent with observation-based low- and moderate-wind slope statistics, and whether its total MSS exhibits a plausible flattening trend under high sea states. The second comparison is a constraint-compliance test rather than a fully independent prediction test because the accepted implementation uses Cox–Munk, Hu/TGRS, and Guérin data to construct the nonlinear MSS target; the subsequent geometry and electromagnetic tests therefore provide the independent evidence for the nonlinear structures.

Let $\widehat{\eta}(\boldsymbol{k})$ denote the Fourier coefficient of a synthesized surface, and let k_a and k_c be the along- and cross-wind wavenumber components defined in Section II. The directional and total MSSs and the anisotropy factor are calculated directly from the resolved spectral coefficients as

$$\begin{aligned} m_a &= \sum_{\boldsymbol{k}} k_a^2 \left|\widehat{\eta}(\boldsymbol{k})\right|^2, \quad & m_c &= \sum_{\boldsymbol{k}} k_c^2 \left|\widehat{\eta}(\boldsymbol{k})\right|^2, \\ m_t &= m_a + m_c, & \gamma &= \left(\frac{m_c}{m_a}\right)^{1/2}. \end{aligned} \tag{34}$$

The primary grid uses $\Delta k = k_p/12$ and resolves nonlinear interactions up to $8k_p$. Independent octave tiles then represent the unresolved short-wave bands up to $k_{\max} = 1000\pi\ \mathrm{rad/m}$. Because the tiles occupy disjoint wavenumber bands, their slope variances are added to the primary-band moments. This multiscale construction includes the gravity–capillary contribution needed for comparison with optical-scale slope statistics without requiring one millimeter-spaced grid over the entire sea-surface domain.

For the low- and moderate-wind experiment, the Cox–Munk reference is evaluated after converting the simulated 10-m wind speed to the 12.5-m wind convention, $U_{12.5} = U_{10}/0.98$. Its directional components are [26]

$$\begin{aligned} m_{a,\mathrm{CM}} &= 3.16 \times 10^{-3} U_{12.5}, \\ m_{c,\mathrm{CM}} &= 3.00 \times 10^{-3} + 1.92 \times 10^{-3} U_{12.5}. \end{aligned} \tag{35}$$

The comparison additionally includes the direction-resolved IASI values reported by Guérin over their available wind interval and the Hu total-MSS parameterization and representative anisotropy summarized in the generalized slope model [27], [28]. These references are retained separately in the plots; their

Placeholder for low- and moderate-wind global statistics

(a) Total MSS of the linear Elfouhaily group, including 5th–95th percentile and interquartile bands, the numerical Elfouhaily integral, Cox–Munk, Hu/TGRS, Guérin IASI, and the empirical fusion target.

(b) Corresponding total MSS of the MSS-constrained Modified-Lie group.

(c) Anisotropy factor γ for both groups and the available reference values.

Fig. 5. Global slope statistics for $U_{10} = 1\text{–}10\ \mathrm{m/s}$ using 20 paired realizations per wind speed. The fusion curve is the constraint target and is not treated as an additional independent observation.

component-wise arithmetic mean is used only as the accepted implementation's low- and moderate-wind constraint target. Denoting this target by $m_{q,\mathrm{tar}}$ and the uncorrected Modified-Lie moment by $m_{q,\mathrm{raw}}$, the desired moment of each realization is

$$m_{q,\mathrm{des}} = m_{q,\mathrm{tar}} \left(\frac{m_{q,\mathrm{raw}}}{m_{q,\mathrm{tar}}} \right)^{\rho}, \qquad q \in \{a, c\}, \tag{36}$$

where $\rho = 0.15$ retains part of the realization-to-realization deviation. The positive angular dressing in Section II is then solved to realize $m_{a,\mathrm{des}}$ and $m_{c,\mathrm{des}}$ without changing Fourier phase. The reported groups are therefore a linear Elfouhaily baseline and an MSS-constrained Modified-Lie surface generated with identical random coefficients.

Wind speed is varied from 1 to $10\ \mathrm{m/s}$ in $1\ \mathrm{m/s}$ increments, with 20 paired realizations at every wind speed. At each condition, the median is used as the central estimate; the 5th–95th percentile interval and interquartile range describe Monte Carlo dispersion. Agreement with a reference curve r_i is summarized by

$$\mathrm{RMSE} = \left[\frac{1}{N_U} \sum_{i=1}^{N_U} \left(\widetilde{m}_{t,i} - r_i \right)^2 \right]^{1/2}, \tag{37}$$

where $\widetilde{m}_{t,i}$ is the median total MSS at the ith wind speed. Directional consistency is assessed independently through the mean absolute deviation of $\widetilde{\gamma}$ from the TGRS reference value.

As shown by the results reserved for Fig. 5, the linear realizations closely reproduce the numerical integral of the implemented Elfouhaily spectrum, with an RMSE of 7.53×10^{-4}. This confirms the self-consistency of the spectral sampling, octave-band accumulation, and MSS calculation. Their total MSS nevertheless increasingly exceeds the Cox–Munk and Guérin references toward the upper end of the interval: the median rises from 0.01846 at $1\ \mathrm{m/s}$ to 0.06156 at $10\ \mathrm{m/s}$, whereas the corresponding Cox–Munk values are 0.00818 and 0.05484. After directional moment contraction, the nonlinear medians become 0.01221 and 0.05547, respectively, and remain monotonic over the complete wind interval.

The quantitative errors exhibit the same behavior. Relative to the linear baseline, the constrained nonlinear group reduces the total-MSS RMSE from 0.00560 to 0.00223 against Cox–Munk, from 0.00523 to 0.00188 against the available Guérin values, and from 0.00383 to 0.00227 against the Hu/TGRS law. Its RMSE relative to the fusion target is 6.28×10^{-4}, compared with 0.00456 for the linear group. The mean absolute anisotropy error relative to the TGRS value decreases from 0.04699 to 0.02678. The constrained γ changes from 0.948 at $1\ \mathrm{m/s}$ to 0.841 at $10\ \mathrm{m/s}$; its elevated low-wind value follows partly from the constant crosswind term in the Cox–Munk law and should therefore be interpreted as a property of the adopted directional constraint rather than as an independently predicted nonlinear-wave effect.

The high-sea-state experiment tests a different statistical behavior and does not extrapolate the Cox–Munk linear law into hurricane-force winds. For $U_{10} \geq 13.3\ \mathrm{m/s}$, the accepted implementation adopts the logarithmic Hu/TGRS branch

$$m_{t,\mathrm{Hu}} = 0.138 \log_{10} U_{10} - 0.084, \tag{38}$$

while the lower part of this experiment uses its adjacent linear branch. The test covers $U_{10} = 10 : 2 : 40\ \mathrm{m/s}$ with 20 paired realizations per condition and retains $\rho = 0.05$. In addition to the linear and constrained groups, the raw Modified-Lie result is reported to expose the effect of the statistical constraint. High-wind behavior is measured by the RMSE to the Hu/TGRS target, the ratio between the mean tail slope above $24\ \mathrm{m/s}$ and the early slope below $16\ \mathrm{m/s}$, and the fraction of successive slope intervals that are concave.

The high-wind results reserved for Fig. 6 show that the raw Modified-Lie correction alone changes the total MSS only slightly and yields an RMSE of 0.00911, compared with 0.00900 for the linear realizations and 0.00757 for the continuous Elfouhaily integral. The constrained group reduces this value to 3.96×10^{-4}. Its median total MSS increases from 0.05455 at $10\ \mathrm{m/s}$ to 0.13754 at $40\ \mathrm{m/s}$, remains strictly monotonic, and is concave over all evaluated finite-difference intervals. Its tail-to-early slope ratio is 0.411, indicating that roughness continues to grow while its wind-speed sensitivity decreases. Together, the two experiments show that the multiscale generator reproduces its prescribed spectral slope energy and that the accepted statistical constraint prevents the nonlinear transform from producing unbounded or incorrectly directed MSS over a broad range of sea states. Because the

Placeholder for high-sea-state MSS validation

(a) Median total MSS for the linear Elfouhaily, raw Modified-Lie, and MSS-constrained Modified-Lie groups, together with the Hu/TGRS law, the integrated Elfouhaily result, and the 5th–95th percentile band of the constrained group.

(b) Finite-difference slope $\mathrm{d(MSS)}/\mathrm{d}U_{10}$ showing the progressive high-wind flattening.

Fig. 6. High-sea-state total MSS and its wind-speed derivative for $U_{10} = 10$–$40\ \mathrm{m/s}$.

observation-based laws participate in that constraint, these results establish global statistical consistency; the independent validity of the localized nonlinear geometry and its radar consequence is examined next.

B. Local Curling-Wave Geometry Validation

The global MSS analysis does not determine whether the localized transformation produces a geometrically credible overturning crest. The local validation therefore considers the front-face angle, horizontal and vertical curl scales, and front–rear asymmetry. Experimental profiles of weak-to-strong plunging breakers reported by Erinin *et al.* provide the primary morphology envelope [29]. Because the model parameters were adjusted using these intervals, the results quantify observation-constrained morphology reproduction rather than an independent prediction of an unseen breaker population.

For each generated event, a center-plane profile is extracted in the propagation direction. Let $\mathcal{P}_c$ be the set of profile points whose curl angle exceeds 8% of its event maximum. The horizontal and vertical curl scales are

$$r_x = \max_{j\in\mathcal{P}_c} u'_j - \min_{j\in\mathcal{P}_c} u'_j, \qquad r_y = \max_{j\in\mathcal{P}_c} z'_j - \min_{j\in\mathcal{P}_c} z'_j. \quad (39)$$

The forward 35% of this support is restricted to the central 15%–85% of its vertical span. When the folded profile contributes points from both branches, only the descending face satisfying $(\mathrm{d}u'/\mathrm{d}u)(\mathrm{d}z'/\mathrm{d}u) < 0$ is retained. If $\boldsymbol{v}_1 = [v_u, v_z]^{\mathrm{T}}$ is the first right singular vector of the centered front-face coordinates, the fitted face angle is

$$\theta_f = \tan^{-1}\left(\frac{|v_z|}{|v_u|}\right). \quad (40)$$

All lengths are normalized by the spectral-peak wavelength $\lambda_p = 2\pi/k_p$. The adopted experimental envelope is $65^\circ \le \theta_f \le 70^\circ$, $0.049 \le r_x/\lambda_p \le 0.069$, and $0.045 \le r_y/\lambda_p \le 0.059$. These intervals describe the selected pre-impact plunging-breaker profiles and are not interpreted as universal breaking-onset thresholds.

The scale experiment uses $U_{10} = 5\ \mathrm{m/s}$, inverse wave age 0.84, and $H_s = 0.35\ \mathrm{m}$. One hundred realizations are generated for both the original shallow-curl parameter domain and the corrected domain; the random sea seed and five curling parameters vary independently within each group. A realization passes only when θ_f, r_x/λ_p, and r_y/λ_p simultaneously lie in their respective intervals. The original group produces no joint pass and has mean values 6.69°, 0.04482, and 0.00210. The corrected group yields $\theta_f = 68.10^\circ \pm 0.35^\circ$, $r_x/\lambda_p = 0.05640 \pm 0.00051$, and $r_y/\lambda_p = 0.05290 \pm 0.00059$, where the reported uncertainties are 95% confidence intervals of the means. Eighty-two of the 100 corrected realizations satisfy all three conditions simultaneously. Thus, the corrected transformation produces the intended mature forward-and-downward curl over a substantial portion of the sampled parameter domain, whereas the previous setting mainly produced a shallow forward displacement.

Front–rear asymmetry is evaluated from the same material profile. Let h_c be the crest elevation above the full-surface mean water level, and let L_f and L_r be the propagation-coordinate distances from the crest to the adjacent forward and rear mean-level crossings. The steepness and asymmetry metrics are

$$\epsilon_f = \frac{h_c}{L_f}, \qquad \epsilon_r = \frac{h_c}{L_r}, \qquad A_{fr} = \frac{\epsilon_f}{\epsilon_r}. \quad (41)$$

The center profile is periodically wrapped about the detected crest so that events near a domain boundary retain both neighboring crossings. Moreover, the crossings are tracked using the undeformed carrier-wave material indices; the plunging jet's own intersection with the mean level is not mistaken for the outer wave boundary. Since a static random height realization has no signed phase velocity, the 0° and 180° propagation orientations are treated as two candidate conditional configurations.

The asymmetry experiment compares 100 paired G0 no-curl, G1 shallow-curl, and G2 conditioned realizations. G2 is retained only when the selected orientation simultaneously yields $65^\circ \le \theta_f \le 70^\circ$ and the auxiliary interval $1.60 \le A_{fr} \le 2.35$. A total of 487 candidate seed–parameter pairs are required to obtain 100 complete pairs, corresponding to an acceptance rate of 20.53%. The median θ_f values are 5.66°, 6.08°, and 67.55° for G0, G1, and G2, respectively. Their median A_{fr} values are 2.085, 2.254, and 1.861, and the probability of a steeper front face, $P(A_{fr} > 1)$, is 0.99, 1.00, and 1.00. The G2 front-angle bootstrap interval is 67.27°–68.03°, while its asymmetry interval is 1.805–1.960.

The two experiments answer complementary questions. The scale test shows that the corrected parameter domain frequently generates curl dimensions and face angles within the laboratory envelope without selecting realizations by their

Placeholder for local curling-wave geometry validation

(a) Joint distribution of front-face angle and normalized horizontal curl scale for the original and corrected parameter domains, with the Erinin envelope.

(b) Corresponding front-face angle and normalized vertical curl scale.

(c) Front-face angle distributions of the paired G0, G1, and conditioned G2 surfaces.

(d) Front–rear asymmetry A_{fr}, including the auxiliary morphology interval and median with 95% bootstrap confidence interval.

Fig. 7. Local geometry assessment of the curling transformation. Scale agreement is evaluated over unscreened Monte Carlo realizations, whereas G2 in the angle–asymmetry comparison is a conditional-generation group with a 20.53% acceptance rate.

measured outcome. The conditioned test shows that the simulator can deliberately produce a prescribed angle–asymmetry family and quantifies the sampling cost required to do so. It does not establish a 100% unconditional prediction rate, nor does the asymmetry result prove that the local rotation alone creates the carrier-wave asymmetry, because the conditioning also selects the propagation orientation. Subject to these boundaries, the generated patches reproduce the principal radar-relevant geometric features: a steep descending front face, finite horizontal and vertical curl scales, front–rear asymmetry, and a locally multivalued projection.

C. Local Electromagnetic Scattering Validation

The geometry test establishes the shape of the curling patch but does not show whether that shape produces a meaningful change in radar-facing surface response. A strictly paired local experiment is therefore performed on the same random sea, detected crest, material window, mesh connectivity, and radar look direction. G0 uses the uncurled vertices and G1 uses the curled vertices; the face indices are selected once in G0 material coordinates and reused in G1. This arrangement isolates changes in facet area, orientation, and visibility from changes in the sampled sea realization or integration footprint.

The current validation code follows the existing echo implementation and assigns visible facet i the single-channel response proxy

$$\widetilde{\sigma}_i = A_i \left[\max\left(\boldsymbol{n}_i \cdot \boldsymbol{s}, 0\right)\right]^2, \tag{42}$$

where A_i, $\boldsymbol{n}_i$, and $\boldsymbol{s}$ are the facet area, upward-oriented normal, and unit direction to the radar. For a fixed material window Ω_w with horizontal area A_w, the unit-area local response is

$$\widetilde{\sigma}_g^0 = \frac{1}{A_w} \sum_{i \in \Omega_w} \widetilde{\sigma}_{i,g}, \qquad g \in \{0, 1\}. \tag{43}$$

The curling-wave enhancement is then

$$G_b = 10 \log_{10} \frac{\widetilde{\sigma}_1^0}{\widetilde{\sigma}_0^0} = \widetilde{\sigma}_{1,\mathrm{dB}}^0 - \widetilde{\sigma}_{0,\mathrm{dB}}^0. \tag{44}$$

Because both groups use the same A_w, this ratio is also equal to the ratio of their locally integrated response proxies. The tilde distinguishes this quantity from a rigorously calibrated, polarization- and frequency-dependent NRCS. The nominal 10-GHz frequency is metadata in this experiment; (42) itself contains no wavelength, dielectric, Bragg, diffraction, or coherent multipath term.

Sixty G0/G1 pairs are generated on $32 \times 32\ \mathrm{m}^2$ surfaces with $\Delta x = \Delta y = 0.05\ \mathrm{m}$, $U_{10} = 5\ \mathrm{m/s}$, and $H_s = 0.35\ \mathrm{m}$. The fixed local footprint is $3.2\ \mathrm{m}$ in the propagation direction and $6.4\ \mathrm{m}$ in the crestwise direction, giving $A_w = 20.48\ \mathrm{m}^2$. The radar is placed in the up-wave direction at a nominal grazing angle of 12°. G1 uses a continuous morphology control $\chi \in [0.15, 1.00]$, sampled by stratification rather than by discrete weak, moderate, and strong groups, and its maximum rotation multiplier is 1.35χ. The response is summarized by all paired G_b values and by medians and interquartile ranges in eight χ bins.

For an external magnitude reference, the fixed HH panel in Fig. 8(a) of Kim and Johnson is digitized [30]. At each LONGTANK stage, the largest of the reported direct, single-bounce, and double-bounce maximum image magnitudes is retained. The median over stages 1–8, -21.65 dB, defines a prebreaking reference, and the derived breaking-stage enhancement for stages 9–16 is

$$G_{b,\mathrm{ref}}(n) = S_{\mathrm{dom}}(n) - \mathrm{median}_{1 \le j \le 8}\, S_{\mathrm{dom}}(j). \tag{45}$$

The extracted values have a median of 6.10 dB, a range of 2.25–7.85 dB, and an estimated digitization uncertainty of ± 1 dB. This is a derived comparison quantity, not an NRCS metric defined by the source. Moreover, the literature abscissa is breaker evolution stage, whereas χ is a model control parameter; the two axes are not mapped to one another. The reference is consequently used as a stage envelope and a distribution-level magnitude comparison, not as a pointwise response curve.

All 60 paired cases yield $G_b > 0$ under the adopted response proxy. The median enhancement is 5.069 dB, with an interquartile interval of 2.312–6.919 dB and a complete range of 0.127–9.046 dB. The model median is 1.031 dB below the literature-derived median, while the complete model interquartile interval lies inside the 2.25–7.85-dB literature stage envelope. This supports agreement in the order of the

Placeholder for local electromagnetic scattering validation

(a) Sixty paired G_b samples versus the continuous model control χ, with eight-bin medians and interquartile ranges; the literature-derived enhancement is shown only as a horizontal stage envelope and median.

(b) Literature and present-model enhancement distributions shown as two categories, including raw samples, medians, interquartile ranges, and the $\pm$1-dB digitization uncertainty of the literature points.

Fig. 8. Paired local scattering response of the curling geometry. The literature breaker stage and the model control χ are not treated as equivalent coordinates.

local enhancement rather than agreement in absolute scattering calibration.

The response also varies smoothly with the model control. The Pearson and Spearman correlations between χ and G_b are 0.9794 and 0.9823, respectively, and a linear summary gives a slope of $10.592\ \mathrm{dB}/\chi$. The median enhancement increases from 0.295 dB in the first χ bin to 8.025 dB in the last. This monotonicity demonstrates numerical control and geometric sensitivity within the adopted facet proxy; it is not expected to reproduce the nonmonotonic LONGTANK stage sequence, where coherent direct, single-bounce, and double-bounce mechanisms exchange dominance. Full-wave diffraction, multipath interference, foam, spray, and polarization-dependent scattering remain outside the local proxy. The present experiment therefore establishes that the validated curling geometry generates a stable unit-area radar-facing response of a literature-consistent magnitude, while absolute NRCS and coherent breaking-wave electromagnetics are not claimed.

D. *Full-Scene Radar Echo and Statistical Validation*

The preceding tests isolate global surface statistics, local curling geometry, and its unit-area scattering consequence. This subsection evaluates whether the simulated full-scene complex echoes reproduce the range–time organization and amplitude statistics of measured sea clutter. The accepted full-scene experiment compares a linear Elfouhaily baseline, the proposed nonlinear background with directionally organized wind waves, and a measured staring-mode record. Localized curling events are not inserted as a separate class in this run because their geometry and local response were evaluated in Sections III-B and III-C; the conclusions below therefore concern the large-area nonlinear wind-wave and statistical echo components.

The measured case provides $U_{10} = 8.32$ m/s, $H_s = 0.839$ m, mean period $T_{01} = 3.327$ s, wind-from direction 342.4°, and wave-from direction 353.0°. The radar operates at 9.4 GHz with HH polarization and an antenna height of 80 m. Its measured header supplies the pulse-repetition frequency, range sampling, first range, staring direction, and available pulse count. Both simulations use 4096 pulses over the matched 1–5-km interval, the same random broadband realization, and the same radar and receiver parameters. The proposed group reallocates 35% of the elevation variance to a directional wind-wave band centered at T_{01} with a directional spreading exponent of 18, then applies the nonlinear Lie/Creamer correction. The resulting significant wave heights are 0.8390 and 0.8409 m for the linear and proposed groups, respectively, so the comparison does not obtain its statistical differences from a material increase in total wave energy.

The time-averaged range profile is defined as

$$\overline{P}(r_b) = \frac{1}{N_p} \sum_{n=1}^{N_p} |y_{n,b}|^2, \tag{46}$$

where $y_{n,b}$ is the complex pulse-compressed echo in pulse n and range cell b. Absolute Linear/Proposed comparisons retain their common physical simulation scale. For structural comparison with the uncalibrated measured analog-to-digital-converter data, each echo is compensated only for the theoretical R^{-4} power loss, or R^{-2} in amplitude, and then normalized by one global group RMS. No empirical per-range normalization is applied, because that operation would suppress the range texture to be evaluated.

The range profile and range–time intensity (RTI) image provide complementary evidence. The absolute profiles and RTIs verify that the simulated power decreases with range under the radar equation, while the structurally compensated RTIs expose slow-time persistence and range-localized bright bands. Over the homogeneous 2.5–3.0-km gate, the proposed mean power is 5.95×10^{-21}, compared with 6.70×10^{-21} for the linear baseline, but its 99.9th-percentile amplitude is 1.267 times the corresponding linear value. Thus, the proposed model produces stronger intermittent events without imposing a higher mean-power scale. Visually, the linear RTI remains predominantly fine-grained, whereas the proposed RTI contains longer-lived range structures similar to those visible in the measured record.

Slow-time and range persistence are quantified from the centered intensity autocorrelation

$$\rho_I(\tau) = \frac{\mathbb{E}\{[I(t) - \mu_I][I(t+\tau) - \mu_I]\}}{\mathbb{E}\{[I(t) - \mu_I]^2\}}, \qquad I(t) = |y(t)|^2. \tag{47}$$

The lag-one intensity correlations are 0.857, 0.963, and 0.947 for Linear, Proposed, and Measured, respectively. At a 100-ms lag, they are 0.0524, 0.4626, and 0.1494, while the integrated correlation times are 0.0218, 0.1264, and 0.0423 s.

Placeholder for full-scene range and RTI validation

(a) Time-averaged Linear and Proposed range profiles on their common absolute simulation scale.

(b)–(d) Linear, Proposed, and Measured RTIs after theoretical R^2 amplitude compensation and one global RMS normalization per group; all panels use a common display range and no empirical per-range envelope removal.

Fig. 9. Full-scene range profiles and RTIs for the matched wind, wave, and radar conditions.

Placeholder for full-scene statistical validation

(a) Slow-time and range intensity autocorrelation of Linear, Proposed, and Measured echoes.

(b) Empirical RMS-normalized amplitude PDFs.

(c) Empirical CDFs.

(d) Empirical CCDFs on a logarithmic probability axis, with Rayleigh, Weibull, log-normal, and K fits shown where required.

Fig. 10. Temporal, spatial, and amplitude-distribution statistics in the fixed 2.5–3.0-km sea-clutter gate.

The proposed return therefore corrects the near-memoryless behavior of the linear baseline but remains more persistent than this measured realization. By contrast, the 5-m range correlations are 0.703, 0.649, and 0.656, showing close agreement between the proposed and measured range continuity. This result supports the presence of coherent slow-time structures but also identifies the proposed texture lifetime as an upper-biased calibration for the present case.

For amplitude-distribution analysis, samples are drawn from the fixed 2.5–3.0-km homogeneous gate after theoretical range compensation. Each group uses a single normalization

$$a = \frac{|y|}{\sqrt{\mathbb{E}|y|^2}}, \tag{48}$$

which preserves spatial power texture within the gate. The empirical cumulative distribution and complementary cumulative distribution are $\widehat{F}(a)$ and $\widehat{\overline{F}}(a) = 1 - \widehat{F}(a)$. Rayleigh, Weibull, log-normal, and K amplitude models are fitted after echo generation. Distribution agreement with the measured samples is evaluated using the two-sample Kolmogorov–Smirnov distance, the one-dimensional Wasserstein distance, and the logarithmic tail error

$$E_{\text{tail}} = \left[\frac{1}{N_t} \sum_{j \in \mathcal{T}} \left(\log_{10} \widehat{\overline{F}}_s(a_j) - \log_{10} \widehat{\overline{F}}_m(a_j) \right)^2 \right]^{1/2}, \tag{49}$$

where $\mathcal{T}$ contains measured tail probabilities from 10^{-4} to 10^{-1}.

The linear amplitudes are nearly Rayleigh: their Weibull shape is 2.004 and the fitted K shape parameter reaches 90.30, effectively the Rayleigh limit. The proposed and measured amplitudes are both heavy tailed. Their Weibull shapes are 1.380 and 1.523, their K shape parameters are 1.095 and 1.741, and their normalized 99.9th-percentile amplitudes are 3.586 and 3.806, respectively, compared with 2.598 for Linear. The probabilities above twice the RMS amplitude are 0.01796, 0.04683, and 0.04430. Among the four parametric fits, Weibull gives the lowest information criterion for the proposed samples, whereas K gives the best fit to the measured samples; accordingly, the comparison is based primarily on the empirical PDF/CDF/CCDF rather than on requiring the two datasets to select the same named distribution.

Direct empirical distances confirm the improvement. Replacing the linear baseline with the proposed model reduces the two-sample KS distance from 0.1254 to 0.0592, the Wasserstein distance from 0.1266 to 0.0380, and the logarithmic CCDF-tail RMSE from 1.7948 to 0.4420. The normalized 99th-percentile error changes from -22.99% to -1.09%, while the error in the probability of exceeding the measured 99th-percentile threshold decreases from -9.66×10^{-3} to -6.63×10^{-4}. The proposed echo thus reproduces the measured marginal distribution and tail probability much more closely than the linear model, although its slow-time persistence is too strong.

These results must be interpreted as a matched-condition reproduction rather than a fully independent generalization test. The fast and slow speckle time scales and the effective range-response bandwidth were estimated from the measured autocorrelation, and the proposed echo additionally uses a unit-mean, correlated Gamma power texture with a nominal shape of one, a 1.8-s time scale, and a 90-m range scale. This compound-Gaussian texture is a resolution-cell wind-wave modulation model applied after pulse compression; it is not produced by the TSC kernel or by localized curling geometry, and it was selected to reproduce the observed persistence and heavy tail without changing mean power. Consequently, the present case establishes end-to-end consistency of the surface-to-echo generator under a calibrated sea state. Verification on held-out acquisitions, wind speeds, and viewing geometries remains necessary before claiming environment-wide statistical generalization.

E. Computational Cost and Dataset Scalability

The preceding validations establish statistical, geometric, local-scattering, and full-scene echo consistency. This final experiment examines whether the complete generator is computationally suitable for producing large radar datasets. One timed sample comprises directional spectral sampling and inverse fast Fourier transformation (IFFT), the nonlinear Lie/Creamer correction and horizontal displacement, equal-energy directional short-wave replacement, localized curling, triangular-facet construction and TSC-based range-bin accumulation, correlated speckle/texture synthesis, and pulse-compression filtering. Optional MAT-file storage is timed separately. This definition follows the actual resolution-cell statistical echo chain used in this work; it must not be interpreted as a full-wave solution or as coherent accumulation over every facet at every pulse.

Let $N_g = N_x N_y$ be the number of surface vertices, $N_f = 2(N_x - 1)(N_y - 1)$ the number of triangular facets, N_p the number of pulses, N_r the number of range cells, and K the pulse-compression kernel length. The measured compute time is decomposed as

$$T_{\text{comp}} = T_{\text{base}} + T_{\text{Lie}} + T_{\text{short}} + T_{\text{curl}} + T_{\text{EM}} + T_{\text{echo}}, \quad (50)$$

where T_{EM} includes mesh construction, facet-wise TSC evaluation, visibility testing, the radar equation, and range-bin accumulation. The end-to-end stored-sample time is $T_{\text{sample}} = T_{\text{comp}} + T_{\text{IO}}$. For the single-geometry realization used in the benchmark, the leading computational order is

$$T_{\text{comp}} = \mathcal{O}(N_g \log N_g) + \mathcal{O}(N_f) + \mathcal{O}(N_p N_r) + \mathcal{O}(N_p N_r K). \quad (51)$$

For a dynamic sequence containing N_s independently updated geometries, the surface-generation and facet terms become $\mathcal{O}[N_s(N_g \log N_g + N_f)]$, whereas the precise echo cost depends on how the geometry snapshots are assigned to the N_p pulses.

The scalability test uses square grids of 128, 256, 512, and 1024 points per axis with fixed $\Delta x = \Delta y = 0.25$ m, corresponding to domains from 32×32 to 256×256 m^2 and from 32,258 to 2,093,058 facets. Thus, this experiment measures increasing scene size at fixed spatial sampling, rather than resolution convergence on a fixed domain. Each case generates a 4096-pulse echo, one warm-up realization is discarded, and ten independent realizations are timed. The test was executed in MATLAB R2025b under 64-bit Windows 11 on a host with eight logical cores. Mean, standard deviation, and median are retained for every stage.

The mean compute times for the four grids are 0.0473, 0.0756, 0.2312, and 0.8884 s, respectively. At the largest grid, surface construction requires 0.5245 s, facet-based electromagnetic processing requires 0.3058 s, and echo formation requires 0.0581 s, accounting for 59.0%, 34.4%, and 6.5% of T_{comp}. Within the surface stage, the Lie correction is the largest nonlinear contribution at 0.2315 s, whereas the localized curl requires only 0.0325 s, or 3.7% of the complete compute time. Increasing the facet count by approximately four from 512^2 to 1024^2 increases T_{comp} by a factor of 3.84. The observed high-end behavior is therefore close to linear in facet count, with the expected FFT logarithmic factor and a pulse-dependent echo cost that is comparatively weakly affected by the surface grid.

To translate the per-sample measurements into dataset-scale cost, the serial estimate for M independent samples is $T_{\text{data}} = M\overline{T}_{\text{sample}}$. A worker-level projection is additionally defined as

$$T_{\text{data}}^{(W)} = \frac{M\overline{T}_{\text{sample}}}{W\eta_W}, \quad (52)$$

where W is the worker count and η_W is an explicitly assumed parallel efficiency. These projections are not measured parallel speedups. For the 1024^2 production grid, the mean I/O time is 0.1689 s and the stored-sample time is 1.0574 s. Consequently, one million stored samples require an estimated 293.71 h, or 12.24 days, in serial. Assuming $\eta_8 = 0.85$ and $\eta_{32} = 0.75$, the corresponding projections are 43.19 and 12.24 h. Because independent realizations have no intersample dependency, this dataset-level parallelism is available without changing the physical model, although actual throughput will also depend on memory capacity and storage bandwidth.

A separate 40×40 m^2 case is used to place the proposed method in the context of published 8–10-GHz calculations. The grid contains 160×160 vertices at the same 0.25-m spacing, 50,562 facets, and 4096 echo pulses. Over ten realizations, the proposed method requires 0.0635 ± 0.0181 s of compute time and 0.0142 s of mean I/O time, yielding 0.0776 s per stored sample. The corresponding serial compute-only estimate for one million realizations is 17.63 h, while the I/O-inclusive estimate is 21.57 h.

For the same nominal 40×40 m^2 area, Li *et al.* report 1902.65 s for an 81-view full-polarization FPFSM sweep and 18,934.26 s for SSA-II at 8 GHz [22]. Dividing only by the 81 reported incidence views gives derived values of 23.49 and 233.76 s per view, corresponding to contextual runtime ratios of approximately 370 and 3683 relative to the measured compute time of this work. Liu *et al.* report 35.81 s for serial spectral sea generation, 15.13 s for four-node MPI generation, and 168,166, 53,914.8, and 25,471.3 s for serial, MPI, and MPI–OpenMP SSA-II calculations, respectively, for a 40-m 8-GHz case [23]. A 10-GHz multi-GPU PO–SBR sea–ship range–Doppler simulation over 300×300 m^2

Placeholder for measured runtime and scalability results

(a) Mean compute time versus facet count for the four fixed-spacing grids, with standard-deviation error bars and the individual surface, EM, and echo contributions.

(b) Stage-wise runtime breakdown for background IFFT, Lie correction, directional short waves, local curl, facet/TSC processing, and statistical echo formation.

(c) Derived dataset wall time versus sample count for serial, eight-worker, and 32-worker execution; I/O-inclusive and compute-only quantities are distinguished.

Fig. 11. Measured computational scaling of the proposed generator and derived dataset-production cost. Grid enlargement increases physical area while retaining a 0.25-m sampling interval.

Placeholder for X-band computational-context comparison

(a) Log-scale runtime of the proposed 9.4-GHz 40-m generator, derived per-view FPFSM and SSA-II values, reported serial/parallel 8-GHz spectral and SSA-II calculations, and the reported 10-GHz GPU sea–ship imaging case; methods are separated by comparison tier.

(b) Contextual runtime normalized by the measured compute time of this work.

(c) Serial dataset-cost projection for the closest 40-m comparison, with reported, derived, and measured quantities explicitly identified.

Fig. 12. Computational context at 8–10 GHz. Runtime ratios summarize the published timing records but do not constitute same-hardware, same-resolution, or same-output speedups.

requires approximately 6480 s per image [24]. These values demonstrate the markedly different throughput regimes of dataset-oriented statistical generation and high-order or imaging-oriented electromagnetic simulation, but they are not strict speedup measurements: the cited studies use different hardware, wavelength-dependent discretization, polarizations, angular samples, scene contents, and output products. In particular, the present TSC stage represents unresolved X-band roughness statistically and does not reproduce the full-wave interactions computed by SSA-II or PO–SBR.

No quantitative CFD runtime bar is included because the collected breaking-wave references do not report a timing case aligned with the present scene, resolution, and radar-output definition; assigning an assumed CFD runtime would therefore create an unverifiable comparison. The efficiency results nevertheless support the intended role of the proposed framework: rapid generation of diverse radar-ready sea-surface samples while retaining the validated nonlinear, directional, and localized curling structures. Its practical advantage follows from using FFT-based global synthesis, facet-wise statistical scattering, and localized prescribed curling instead of resolving breaking-wave hydrodynamics and full-wave electromagnetic interactions for every realization. This advantage is obtained through a deliberate fidelity–throughput tradeoff; the method is a scalable data generator rather than a replacement for CFD or full-wave solvers when detailed fluid evolution, diffraction, or coherent multipath fields are the quantities of interest.

IV. Conclusion

This letter presented a computationally efficient framework for generating radar-oriented nonlinear sea surfaces across multiple environmental conditions. The framework begins with an Elfouhaily spectral realization, applies a nonlinear Lie-transform correction to the background surface, replaces a prescribed short-wave band with equal-energy directionally organized wind waves, and introduces localized curling geometry at selected steep crests. A facet-based two-scale scattering calculation and a statistical complex-echo model then connect the multiscale geometry to radar observables. This hierarchical construction retains explicit control over sea state, propagation direction, short-wave organization, and breaker morphology while avoiding the cost of resolving the complete hydrodynamic and full-wave electromagnetic evolution of every scene.

Validation was performed progressively from global statistics to local geometry, local scattering, full-scene observations, and computational practicality. The constrained nonlinear surfaces reduced the low-wind total-MSS RMSE from 0.00560 to 0.00223 relative to Cox–Munk and from 0.00523 to 0.00188 relative to the Guérin reference, while preserving monotonic and saturating high-wind behavior. The conditioned curling waves reproduced the observation-constrained front-face-angle and front–rear-asymmetry ranges, and their unit-area radar-facing response produced a median enhancement of 5.069 dB, within the 2.25–7.85-dB literature-derived breaking-stage envelope. At the full-scene level, the proposed echoes improved the two-sample KS distance from 0.1254 to 0.0592, the Wasserstein distance from 0.1266 to 0.0380, and the logarithmic CCDF-tail RMSE from 1.7948 to 0.4420 relative to the linear baseline. The largest tested 1024×1024 grid required 0.8884 s of computation, and a stored 40×40 m^2 realization required 0.0776 s on average, corresponding to an estimated 21.57 h for one million serial samples.

The present results support the proposed method as a controllable and scalable generator for large radar-oriented synthetic datasets, but they do not constitute a complete hydrodynamic or coherent full-wave description of breaking seas. The local electromagnetic experiment validates a facet-based response proxy rather than absolute breaker NRCS, and the accepted full-scene comparison does not insert localized curling events as a separate echo class. Foam, spray, entrained air, diffraction, multipath interference, and polarization-dependent breaker dynamics also remain outside the current formulation, while the simulated slow-time persistence is stronger than that of the measured record. Future work will therefore target held-out sea states and radar geometries, time-evolving breaker populations, selective higher-fidelity scattering for the most nonlinear regions, and downstream learning experiments that quantify how the generated diversity improves radar detection and recognition.

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